



Name: \_\_\_\_\_

Date: \_\_\_\_\_

## LIGHT

Solve the following problems. Write your answer on the line.

Score: \_\_\_\_\_ / 20

Grade 6 | Science | Sheet 1

|                                                                                                                                                      |                                                                          |                                                                         |                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1) Light travels in ____ lines.<br>_____                                                                                                             | 2) Speed of light is ____ m/s (approx).<br>_____                         | 3) Reflection follows the law of ____.<br>_____                         | 4) Angle of incidence = angle of ____.<br>_____                     |
| 5) Refraction is bending of light.<br>_____                                                                                                          | 6) A prism splits light into ____.<br>_____                              | 7) The colors of the visible spectrum are ____.<br>_____                | 8) A convex lens ____ (brings together).<br>_____                   |
| 9) Concave lens diverges (spreads apart).<br>_____                                                                                                   | 10) A plane mirror reflects ____.<br>_____                               | 11) Plane mirror forms a ____ image.<br>_____                           | 12) Virtual image cannot be formed on ____.<br>_____                |
| 13) Opaque objects do not allow light to pass through.<br>_____                                                                                      | 14) Transparent objects allow light to pass through completely.<br>_____ | 15) Translucent objects allow light to pass through partially.<br>_____ | 16) Shadows are formed when light is blocked by an object.<br>_____ |
| 17) Total internal reflection occurs when light travels from a denser medium to a rarer medium at an angle greater than the critical angle.<br>_____ | 18) Optical fibers use ____ to transmit light signals.<br>_____          | 19) Human eye has a ____ lens.<br>_____                                 | 20) Retina converts light into electrical signals.<br>_____         |



Name: \_\_\_\_\_

Date: \_\_\_\_\_

## ANSWER KEY

### Light -- Solutions

Grade 6 | Science | Sheet 1

|                                                                                                    |                                                                                                         |                                                                                                      |                                                                    |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| 1) Light travels in ____ lines.<br><b>straight</b>                                                 | 2) Speed of light is ____ m/s (approx)<br><b>300,000,000 / <math>3 \times 10^8</math></b>               | 3) Reflection follows the law of ____<br><b>reflection</b>                                           | 4) Angle of incidence = angle of ____.<br><b>reflection</b>        |
| 5) Refraction is bending of light when it changes ____.<br><b>medium</b>                           | 6) A prism splits light into ____.<br><b>spectrum/rainbow colors</b>                                    | 7) The colors of the visible spectrum are ____.<br><b>Violet Indigo Blue Green Yellow Orange Red</b> | 8) A convex lens ____ (brings together) light.<br><b>converges</b> |
| 9) Concave lens diverges (spreads out) ____.<br><b>light</b>                                       | 10) A ____ mirror reflects ____.<br><b>plane light</b>                                                  | 11) Plane mirror forms a ____ image.<br><b>virtual</b>                                               | 12) Virtual image cannot be formed on ____.<br><b>screen</b>       |
| 13) Opaque objects do not allow light to ____.<br><b>pass through</b>                              | 14) Transparent objects allow light to ____.<br><b>pass through</b>                                     | 15) Translucent objects allow ____ light to pass.<br><b>partial</b>                                  | 16) Shadows are formed when light is ____.<br><b>blocked</b>       |
| 17) Total internal reflection occurs when the angle of incidence is ____.<br><b>critical angle</b> | 18) ____ is the angle of incidence greater than the critical angle.<br><b>total internal reflection</b> | 19) Human eye has a ____ lens.<br><b>convex</b>                                                      | 20) Retina converts light into ____.<br><b>nerve signals</b>       |